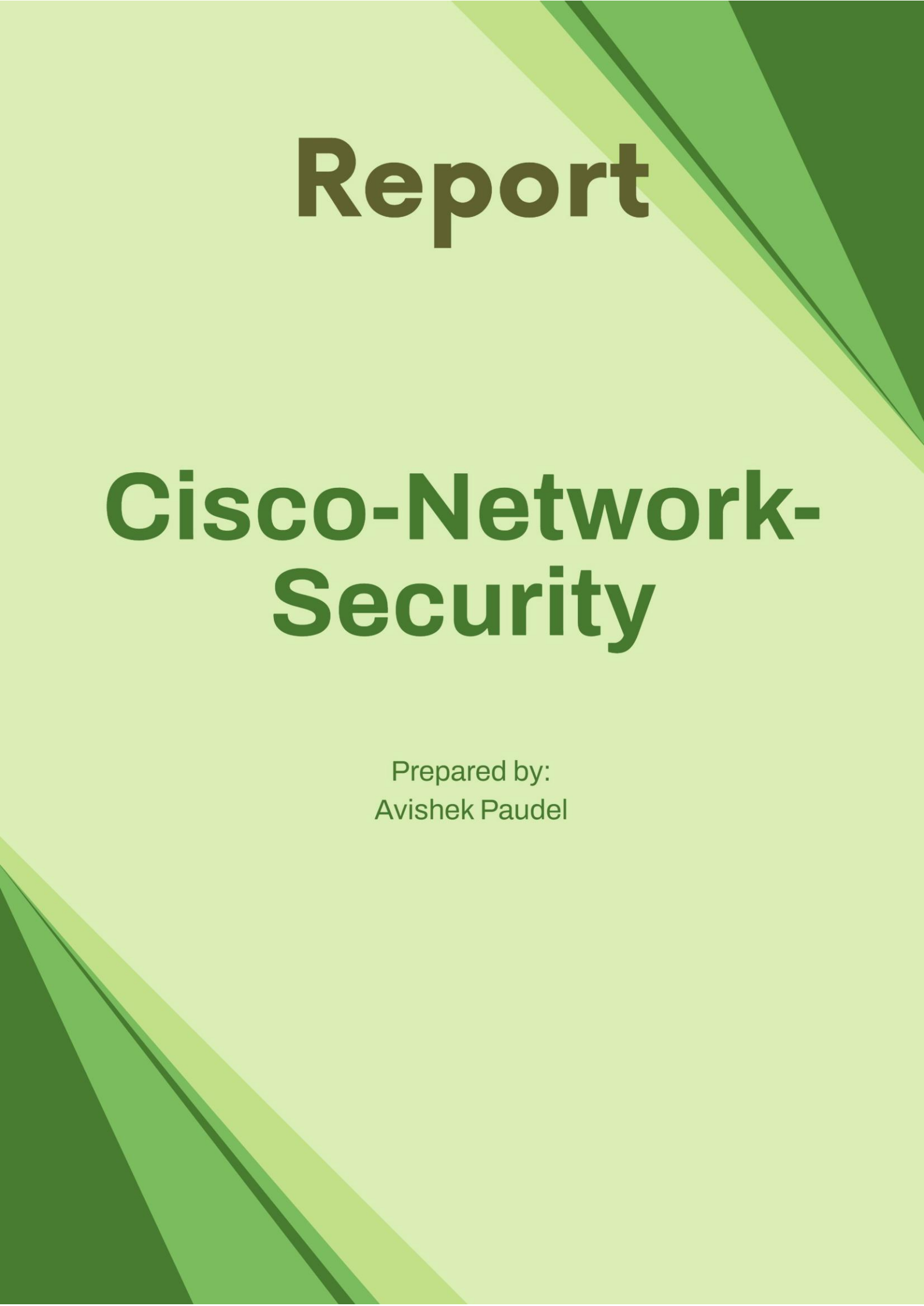


The background of the slide features abstract, overlapping geometric shapes in various shades of green, creating a modern and dynamic visual effect.

Report

Cisco-Network-Security

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Security Implementation Commentary

Introduction

This report provided a commentary on the secure network that was designed and implemented for -the tow Carlisle Sites which are Carlisle Branch (Site-A) and Carlisle HQ (Site-B). The simulation was built by using the Cisco Packet Tracer which covers a range of security measures that were applied across both sites to ensure the network is protected from both internal and external threats.

The network was built in three stages. The first stage involved creating the basic topology with VLANs, OSPF routing and DHCP servers. The second stage added internal security features which includes devices hardening, firewall ACLs, NAT/PAT and OSPF authentication. The third stage is connected the two sites security by using a GRE tunnel protected by IPSec VPN encryption. Each decision is made through the implementation which are explained in detail below.

Task 5 provides the five security policy rules to help the organisation maintain consistent network security to move forward.

Device Access credentials

The following credentials are required to access the network devices within the Packet Tracer simulation files. These credentials must be used by the marker to access any verify device configuration.

Border Routers:

Device	Username	Password
BorderRouter-A	Admin	Adm1nP@ss#1
BorderRouter-B	Admin	Adm1nP@ss#2

Core switches:

Device	Username	Password
CoreSwitch-A	Admin	Adm1nP@ss#3

CoreSwitch-B	Admin	Adm1nP@ss#4
--------------	-------	-------------

AccessSwitches:

Device	Username	Password
AccessSwitch-A1	Admin	Adm1nP@ss#9
AccessSwitch-A2	Admin	Adm1nP@ss#10
AccessSwitch-B1	Admin	Adm1nP@ss#11
AccessSwitch-B2	Admin	Adm1nP@ss#12

Switches

Device	Username	Password
Management Switch-A	Admin	Adm1nP@ss#5
Management Switch-B	Admin	Adm1nP@ss#6
Internal Switch-A	Admin	Adm1nP@ss#7
Internal Switch-B	Admin	Adm1nP@ss#8

Note: All end devices including PCs, Servers, Laptops require no login credentials and it can be accessed directly by clicking the device in Packer Tracer.

Choice of IP Version

IPv4 was selected for this network by using the private address range 10.0.0.0/8 as required by the assignment. The decision to use IPv4 rather than IPv6 came down to a few practical reasons.

The preconfigured ISP infrastructure provided in the starter file already uses IPv4 public addressing with ISP-1 using 1.0.0.0/28 and ISP-2 using 3.0.0.0/28. Using IPv6 internally while the ISP links are IPv4 would have created unnecessary complexity with dual stack configuration and translation mechanism. It would add little benefit in this simulated environment. IPv4 also works significantly better with Cisco Packet Tracer when it comes to

IPSec and GRE tunnel configuration. Packet Tracer also known its limitation with IPv6 support for these features. Therefore, using IPv4 throughout ensured a more stable and reliable simulation.

Another important reason is that IPv4 with NAT and PAT provides an additional security layer by completely hiding the internal private address scheme from anyone on the public internet. This means even if an attacker is monitoring traffic on the ISP links they cannot determine the internal network structure.

The internal address allocated was planned as follows:

Subnet	Purpose
10.10.10.0/24	Site A – VLAN 10 Management
10.10.20.0/24	Site B – VLAN 20 Staff
10.10.30.0/24	Site A – VLAN 30 Servers
10.20.10.0/24	Site B – VLAN 10 Management
10.20.20.0/24	Site B – VLAN 20 Staff
10.20.30.0/24	Site B – VLAN 30 Servers
10.10.0.0/30	Site A – Router to Core Switch Link
10.20.1.0/30	Site B – Router to Core Switch Link
10.0.0.0/30	GRE Tunnel between Sites

Implementation of Security Features

Basic Device Security

One of the first thing applied to the network was basis device security across all 12 devices such as both border routers, both core switches, both management switches, both internal server switches and all four access switches across the two sites.

The **service password-encryption** command was enabled on every device. This which ensure that my password stored in the device configuration is automatically encrypted. It means that if someone were to view the running config, they would not be able to read any passwords in plain text.

Each device was given a unique enable secret password to protect access to privileged EXEC mode. By using a unique passwords per device is considered good security practice because if one device is compromised then the attacker cannot use the same password to access other devices on the network. The passwords followed a structure format such as **Adm1nP@ss#1** for Border Router- A through to the **Adm1nP@ss#12** for Accessswitch-B2. By using a mix of uppercase, lowercase, numbers and special character to make them strong.

Console port security was also applied to every device by setting a password on the **line console** interface. This also protect physical access to the device. So, even if someone gains the physical entry on the building and plugs directly into a device, they are still prompted for a password.

A warning banner was also configured on all device by using the **banner motd** command. The banner read “Unauthorized access is prohibited. Disconnected immediately”. This serves as a legal determine and inform anyone connecting to the device that unauthorised access is not permitted.

SSH version 2 was configured on the both border routers by generating a 1024-bit RSA encryption key and running the **ip ssh version 2** command. The VTY line were also configured with **transport input ssh** and **login local** to ensure only SSH connection are accepted and that local user credentials must be provided. Telnet was also completely blocked as it sends all data including passwords in plain text across the network which makes it a significant security risk.

VLAN Segmentation

Both sites use three VLANs to logically separate different types of network traffic. VLAN 10 carries management traffic, VLAN 20 carries staff traffic and VLAN 30 carries server traffic. This segmentation means that staff devices only VLAN 20 cannot directly reach server infrastructure on VLAN 30 without the traffic being routed through the Core Switch where access control can be applied.

VLAN Number	VLAN Name	Purpose
10	Management	Network device management
20	Staff	Staff PCs

30	Servers	DHCP and Web Servers
----	---------	----------------------

The Core Switches at both sites are configured as Layer 3 devices handling inter-VLAN routing. Trunk links carrying all the three VLANs connect to the Core Switches to the access layer switches and DHCP helper addresses are also configured on each VLAN interface to forward DHCP request to the centralised DHCP servers located in the server VLAN.

OSPF MD5 Authentication

OSPF was chosen as the routing protocol for both sites as it is a standards-based-link-state protocol well suited to enterprise network. To prevent from unauthorised router from injecting false routing information into the network MD5 authentication was also configured on all OSPF enabled interfaces.

The commands “**ip ospf message-digest-key 1 md5 OSPF#LAN#Key1** on Site A and **OSPF#LAN#Key** on Site B” were applied to the relevant interface on Border Router-A, Border Router-B, and the both core switches. The MD5 key must need to match exactly on the both ends of every OSPF adjacency which means any rouge router connection to the network would be unable to form an OSPF neighbourhood and also cannot advertise fake routes. Moreover, without this protection an attacker could connect to the network and redirect traffic through their own equipment and also potentially intercepting sensitive data.

Reasoning Behind Firewall Type and NAT/PAT Deployment

Firewall Decision

A packet filtering firewall was implemented using extended names ACLs on both Border Router rather than a Stateful Zone-Based Firewall. This decision was made because the ISR 4331 routers support extended ACLs natively and Cisco Packet Tracer does not fully support ASA stateful firewall simulation. The packet filter approach still provides a solid perimeter security for this network.

The ACLs names FIREWALL-IN-A and FIREWALL-IN-B were applied to the external facing interface which is G0/0/0 in the inbound on the both routers. This means all the traffic arriving from the internal is inspected against the rules before being allowed into the network. This

rule is followed a default deny approach which means anything not explicitly permitted is automatically blocked. The specific rules applied were:

- Permit established TCP sessions this allows return traffic for connection that were initiated from inside the network such as, web browsing responses.
- Permit ICMP echo-reply this allows ping responses to come back in while blocking unsolicited ping which request from the outside.
- Permit ESP this allows IPSec encrypted VPN traffic to pass through.
- Permit GRE this helps to allow the GRE tunnel traffic between the two sites.
- Permit UDP port 500 this helps to allow NAT traversal for the VPN
- Deny all other IP traffic this is the final catch all rule that blocks everything which is not matched above.

NAT/Pat Decisions

NAT with PAT overload was also deployed on the both Border Routers. The inside interface was set to G0/0/1 which is facing the internal network and for the outside interface was set to G0/0/0 which is facing the ISP. PAT was chosen over standard NAT because in both sites we have multiple devices that need internal access but only a single public IP address is available on each site's WAN link.

The Border Routers are the correct and only logical place to deploy NAT because they sit at the boundary between the private internal network and the public internet. By placing NAT on any other device such as a Core Switch which would be incorrect as translation need to happen at the exact point where private and public addressing meet. NAT also provides security benefits as it helps to ensure internal IP addresses are never visible to external parties. It means making it harder for attackers to map the internal network.

Reasoning Behind VPN Parameters and GRE Tunnels

GRE Tunnel

A GRE tunnel was created between Border Router-A and the Border Router-B to establish a logical point-to-point connection between the two Carlisle sites over the public internet. GRE was also selected because it is a flexible tunnelling protocol that can be encapsulate many different types of traffic and also helps to allows routing protocols to run across the tunnel as

if the two sites were directly connected. The tunnel interface was given IP address from the 10.0.0.0/30 subnet with Router-A assigned 10.0.0.1 and Router-B assigned 10.0.0.2

Moreover, GRE does not provide any encryption or security because it simply wraps packets in a new IP header to route them across the internet. This is why IPSec was added on top of the GRE tunnel to provide the necessary encryption and authentication.

IPSec VPN Parameters

The IPSec configuration was carefully chosen to provide the strong security for inter-site communication. The parameters selected were:

Parameter	Value Chosen	Reason
Encryption Algorithm	AES-256	It is currently considered as unbreakable and used by governments for classified data.
Authentication Method	Pre-shared key	It is straightforward to implement and suitable for this two-site setup.
Hashing Algorithm	SHA-HMAC	It helps to provide strong data integrity verification.
IKE Group	Group 2 (1024-bit Diffie-Hellman)	Its secure key exchange supported by Packet Tracer.
SA Lifetime	3600 seconds	Standard one hour rekey interval to limit exposure if a key is compromised.

AES-256 was specifically chosen as the encryption algorithm because it offers the highest level of security which is currently available in symmetric encryption. It is also a significant stronger than older algorithms such as DES or 3DES which are now consider as outdated and vulnerable.

Pre-shared key authentication was used because it is the most practical approach in Packet Tracer. In a real production environment certified based authentication use a Public Key

infrastructure which would be preferable as it scales better and does not require the same secret to be manually configured on the both sites.

The combination of GRE and IPSec gives the network the best of both worlds. GRE provides the tunnel flexibility needed to carry by routing protocol traffic between sites while IPSec provides the encryption and authentication needed to secure that traffic as it crosses the public internet. Additionally, without IPSec all the traffic passing through the GRE tunnel would be transmitted in plain text and could be intercepted by anyone with access to the network path between the two ISPs.

Security Policy Document Extract

Rules 1 – Password and Authentication Policy

Every device on the network needs to have its own unique password. Sharing passwords between devices is not acceptable because if one device gets compromised an attacker could use the same credentials to access everything else. Passwords need to be at least 12 character and also should mix letters, numbers and symbols which make them harder to guess. Passwords should never be left a default and should never be blank too. All passwords stored on device which must be encrypted in the configuration. This is done using the enable secret command and service password-encryption which means even if someone views the config file they cannot read the actual password.

When it comes to remote access SSH version 2 is the only acceptable method. Telnet send everything including passwords as plain text across the network which is a serious risk and should never be used. Passwords should need be updated every three months and if anyone thinks their login details have been leaked then they need to report it straight away so it can be sorted within 24 hours.

Rule 2 – Firewall and Traffic Filtering Policy

The default position for all traffic coming in from the internet should need to be block it unless there is a specific reason to allow it. This is something called a default deny approach and it much safer than trying to block only the bad traffic while allowing everything else.

The only traffic that should be allowed in from outside is traffic that is returning in response to a request made from inside the network, ping replies, VPN traffic and the traffic needed it keep the VPN connection alive. Everything else should need to be dropped because any changes to the firewall rules need to go through the IT security manager first which means nobody should be adding or removing rules without any approval. The firewall rules should be looked at every three months to make sure that nothing unnecessary has crept in over time.

Rules 3 – Network Access and VLAN Segmentation Policy

The network at both sited is split into three sperate VLANs. VLAN 10 is for management, VLAN 20 is for staff and VLAN 30 is for servers. This separation exists for a reason which mean it stops staff devices from being able to directly reach server or management system without going through the controlled path.

Staff laptops and phones should only ever be on VLAN 20, Management task like SSHing into a router or switch should only ever happen from VLAN 10. Server access from staff device should also go through the proper routing with access controls in place rather than being direct.

Guest Wi-Fi must need to be kept completely separate from everything internal. A guest connecting to the wireless network should have no way of reaching any internal company resources. If any device is found plugged into the wrong VLAN then it should be disconnected and looked into straight away.

Rule 4 – VPN and Inter-Site Communication Policy

Any data travelling between the Branch site and the HQ site over the internet they must be encrypted. Sending sensitive company information between the sites without using the VPN is not allowed under any circumstances. The VPN must use AES-256 as the minimum

encryption standard. It means anything weaker should not be accepted without a very good reason and sign-off from the security manager.

The VPN key must need to be treated carefully. It should not be shared with people who do not need it, should not be written on a sticky note or stored somewhere unsecured as well as should definitely not be sent over email. The key should be changed every six months as a routine measure and immediately if there is any doubt it has been seen by the wrong person. Looking ahead, the company should need to move to certificate based authentication when possible as it is more scalable and removes the risks that come with managing a shared secret.

Rule 5 – Routing Protocol Security Policy

The routers on the network use OSPF to share routing information with each other. Without proper protection a rogue device could be connect to the network and start sending fake routing updates, potentially redirecting traffic through a malicious device. To stop this MD5 authentication is used on all the OSPF connections so only the trusted devices can participate in routing.

The MD5 key is used for OSPF which must be kept confidential and only shared with network administrators who is genuinely need it. If an admin leaves the company the key needs to be changed straight away. No new router or layer three switch should be allowed to join the OSPF process without the network administrator approving it first. The routing tables on both border routers should be checked regularly mean in weekly to make sure no unexpected routes have shown up that could indicate something suspicious is happening on the network.

Conclusion

This report has provided a detailed explanation of all the security features implemented across the two Carlisle sites. The network was built with a defence in depth philosophy which means multiple layers of security were applied at different points rather than relying on a single protection mechanism.

Device hardening was also applied to all 12 networks device which helps to prevent unauthorised access. Firewall ACLs on both Border Router protect the network perimeter by blocking the unsolicited inbound traffic. NAT and PAT hide the internal address scheme and

allow multiple devices to share a single public IP. OSPF MD5 authentication also prevents rogue from interfering with routing. Finally, the GRE tunnel protected by IPSec AES-256 encryption helps to ensure all traffic between the two sites is securely encrypted as it travels across the public internet.

Together these features form a comprehensive security implementation that protects the organisation at the device level, the network perimeter level as well as the inter-site communication level.

The five security policy rules in Task 5 ensure these technical controls are supported by clear organisation guidelines which provides the company with both a secure network and the government framework which is needed to maintain it.

Appendix

Appendix	Screenshots
Appendix A – Task 1 Topology	Task1_Topology_Overview
Appendix B – Task 1 Verification	RouterA_01_Interface_Brief, RouterA_02_IP_Route, RouterA_03_OSPF_Neighbor, RouterB_01_Interface_Brief, RouterB_02_IP_Route, SwitchA_01_VLAN_Brief
Appendix C – Task 2 Topology	Task2_Topology_Overview
Appendix D – Task 2 Security Verification	RouterA_Security_Login, RouterB_Security_Login, RouterA_04_ACL_Firewall, RouterB_04_ACL_Firewall, RouterA_05_NAT_Stats, RouterB_05_NAT_Stats, RouterA_06_OSPF_MD5
Appendix E – Task 3 Topology	Task3_Topology_Overview
Appendix F – Task 3 VPN Verification	RouterA_07_GRE_Tunnel, RouterB_07_GRE_Tunnel, RouterA_08_VPN_Config, RouterB_08_VPN_Config

Appendix A – Task 1 Topology

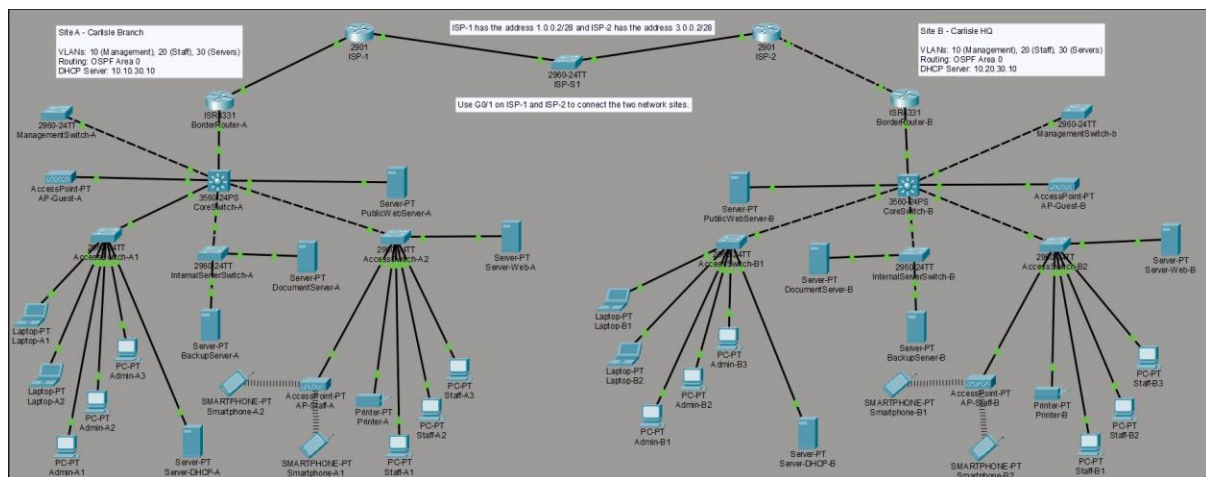


Figure 1: Task1 Topology Overview

Appendix B – Task 1 Verification:

```
BorderRouter-A#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0 1.0.0.1         YES manual up          up
GigabitEthernet0/0/1 10.10.0.2       YES manual up          up
GigabitEthernet0/0/2 unassigned      YES unset  administratively down down
Loopback0           10.10.255.1     YES manual up          up
Vlan1                unassigned      YES unset  administratively down down
BorderRouter-A#
BorderRouter-A#
```

Figure 2: RouterA-Interface

```
BorderRouter-A#show ip ospf neighbor
|
Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          1     FULL/DR         00:00:32    10.10.0.1      GigabitEthernet0/0/1
BorderRouter-A#
```

Figure 3: RouterA-OSPF-Neighbor

```
BorderRouter-A#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 1.0.0.2 to network 0.0.0.0

    1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.0.0.0/28 is directly connected, GigabitEthernet0/0/0
L       1.0.0.1/32 is directly connected, GigabitEthernet0/0/0
    10.0.0.0/8 is variably subnetted, 7 subnets, 4 masks
C       10.10.0.0/30 is directly connected, GigabitEthernet0/0/1
L       10.10.0.2/32 is directly connected, GigabitEthernet0/0/1
O       10.10.10.0/24 [110/2] via 10.10.0.1, 00:02:57, GigabitEthernet0/0/1
O       10.10.20.0/24 [110/2] via 10.10.0.1, 00:02:57, GigabitEthernet0/0/1
O       10.10.30.0/24 [110/2] via 10.10.0.1, 00:02:57, GigabitEthernet0/0/1
C       10.10.255.1/32 is directly connected, Loopback0
S       10.20.0.0/16 [1/0] via 1.0.0.2
S*    0.0.0.0/0 [1/0] via 1.0.0.2

BorderRouter-A#
```

Figure 4: Router-A-IP-Route

```

BorderRouter-B>enable
BorderRouter-B#show ip interface brief
Interface          IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0/0 3.0.0.1        YES manual up            up
GigabitEthernet0/0/1 10.20.0.2       YES manual up            up
GigabitEthernet0/0/2 unassigned      YES unset  administratively down down
Loopback0           10.20.255.1    YES manual up            up
Vlan1                unassigned      YES unset  administratively down down
BorderRouter-B#

```

Figure 5: RouterB-Interface-Brief

```

BorderRouter-B#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 3.0.0.2 to network 0.0.0.0

    3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       3.0.0.0/28 is directly connected, GigabitEthernet0/0/0
L       3.0.0.1/32 is directly connected, GigabitEthernet0/0/0
    10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S       10.10.0.0/16 [1/0] via 3.0.0.2
C       10.20.1.0/30 is directly connected, GigabitEthernet0/0/1
L       10.20.1.2/32 is directly connected, GigabitEthernet0/0/1
C       10.20.255.0/30 is directly connected, Loopback0
L       10.20.255.1/32 is directly connected, Loopback0
S*    0.0.0.0/0 [1/0] via 3.0.0.2

BorderRouter-B#

```

Figure 6: RouterB-IP-Route

```

CoreSwitch-A# show vlan brief

VLAN Name                Status      Ports
-----
1    default                active      Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/2
10   Management              active
20   Staff                   active
30   Servers                 active
99   Native-Unused           active
1002 fddi-default            active
1003 token-ring-default      active
1004 fddinet-default         active
1005 trnet-default           active
CoreSwitch-A#

```

Figure 7: CoreSwitch-A-VLAN-Brief

Appendix C – Task 2 Topology

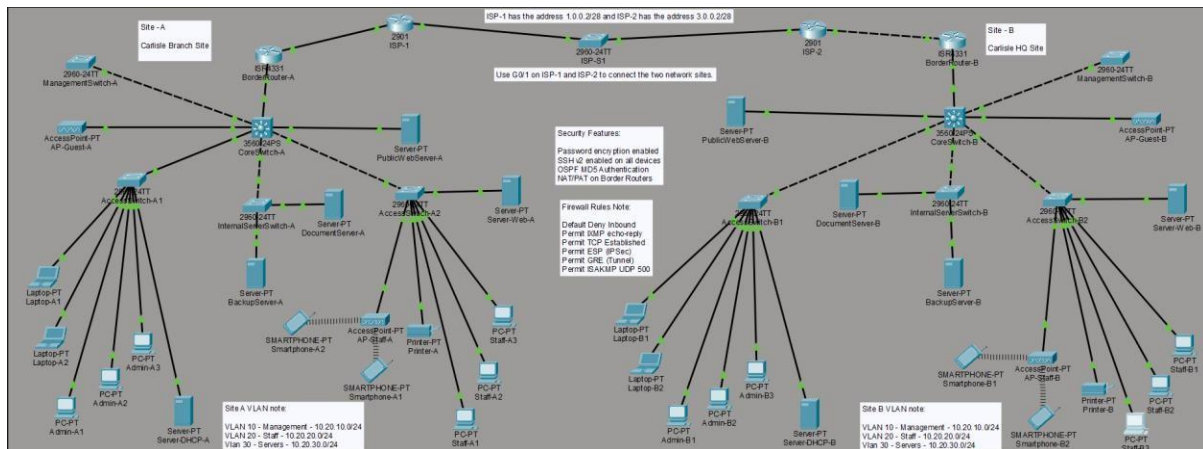


Figure 8: Task2-Topology-Overview

Appendix D – Task 2 Security Verification

```
Unauthorized access is prohibited. Disconnect immediately.

User Access Verification

Password:

BorderRouter-A>enable
Password:
BorderRouter-A#
```

Figure 9: RouterA-Security-Login

```
Press RETURN to get started!

Unauthorized access is prohibited. Disconnect immediately.

User Access Verification

Password:

BorderRouter-B>enable
Password:
```

Figure 10: RouterB-Security-Login

```

BorderRouter-A#show ip access-lists
Standard IP access list NAT-TRAFFIC-A
 10 permit 10.10.0.0 0.0.255.255
Extended IP access list FIREWALL-IN-A
 10 permit tcp any 10.10.0.0 0.0.255.255 established
 20 permit icmp any any echo-reply
 30 permit esp any any
 40 permit gre any any
 50 permit udp any any eq isakmp
 60 permit udp any any eq non500-isakmp
 70 deny ip any any

BorderRouter-A#

```

Figure 11: RouterA-ACL-Firewall

```

BorderRouter-B#show ip access-lists
Standard IP access list NAT-TRAFFIC-B
 10 permit 10.20.0.0 0.0.255.255
Extended IP access list FIREWALL-IN-B
 10 permit tcp any 10.20.0.0 0.0.255.255 established
 20 permit icmp any any echo-reply
 30 permit esp any any
 40 permit gre any any
 50 permit udp any any eq isakmp
 60 permit udp any any eq non500-isakmp
 70 deny ip any any

BorderRouter-B#

```

Figure 12: RouterB-ACL-Firewall

```

BorderRouter-A#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/0/0
Inside Interfaces: GigabitEthernet0/0/1
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
BorderRouter-A#
BorderRouter-A#

```

Figure 13: RouterA-NAT-Stats

```

BorderRouter-B#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/0/0
Inside Interfaces: GigabitEthernet0/0/1
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
BorderRouter-B#

```

Figure 14: RouterB-NAT-Stats

```
BorderRouter-A#show ip ospf interface GigabitEthernet0/0/1

GigabitEthernet0/0/1 is up, line protocol is up
Internet address is 10.10.0.2/30, Area 0
Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 2
Designated Router (ID) 1.1.1.1, Interface address 10.10.0.2
Backup Designated Router (ID) 3.3.3.3, Interface address 10.10.0.1
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:08
Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 3.3.3.3  (Backup Designated Router)
Suppress hello for 0 neighbor(s)
Message digest authentication enabled
  Youngest key id is 1
BorderRouter-A#
```

Figure 15: RouterA-OSPF-MD5

Appendix E – Task 3 Topology

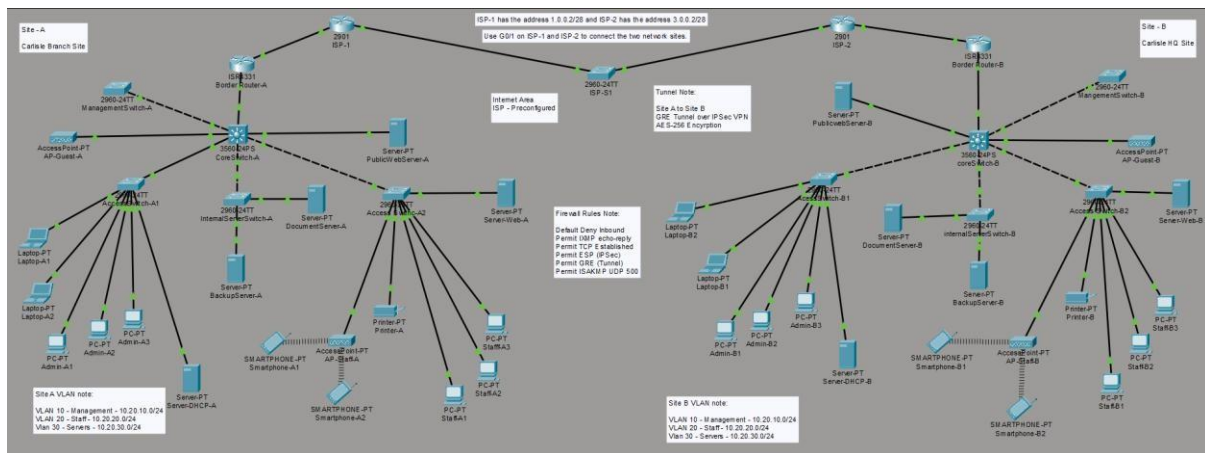


Figure 16: Task3-Topology-Overview

Appendix F – Task 3 VPN Verification

```
BorderRouter-A#show interfaces tunnel0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 10.0.0.1/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 203.0.113.1 (GigabitEthernet0/0/0), destination 3.0.0.1
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out

BorderRouter-A#
```

Figure 17: RouterA-GRE-Tunnel

```

BorderRouter-B#show interfaces tunnel0
Tunnel0 is up, line protocol is down (disabled)
  Hardware is Tunnel
  Internet address is 10.0.0.2/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 203.0.113.5 (GigabitEthernet0/0/0), destination 1.0.0.1
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out

```

Figure 18: RouterB-GRE-Tunnel

```

BorderRouter-A#show crypto map
Crypto Map CMAP-A 10 ipsec-isakmp
  Peer = 3.0.0.1
  Extended IP access list VPN-TRAFFIC-A
    access-list VPN-TRAFFIC-A permit gre host 1.0.0.1 host 3.0.0.1
  Current peer: 3.0.0.1
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (Y/N): N
  Transform sets={
    VPN-SET,
  }
  Interfaces using crypto map CMAP-A:
    GigabitEthernet0/0/0
BorderRouter-A#

```

Figure 19: RouterA-VPN-Config

```
BorderRouter-B#show crypto map
Crypto Map CMAP-B 10 ipsec-isakmp
  Peer = 203.0.113.2
  Extended IP access list VPN-TRAFFIC-B
    access-list VPN-TRAFFIC-B permit gre host 203.0.113.6 host 203.0.113.2
  Current peer: 203.0.113.2
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (Y/N): N
  Transform sets={
    VPN-SET,
  }
  Interfaces using crypto map CMAP-B:
    GigabitEthernet0/0/0

BorderRouter-B#
```

Figure 20: RouterB-VPN-Config